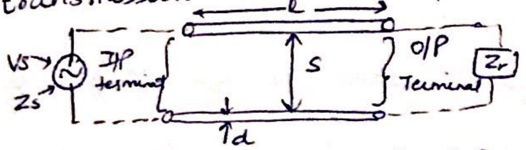
UNIT- IV
Transmission Lines

Transmission lines are a means to convey electrical signals or power b/w two points seperated appreciably in distance. It is a specific example of a linear passive two port n/w of a ~~ckt~~ ckt. theory. Simplest form of transmission lines is a pair of parallel wire insulated from each other.

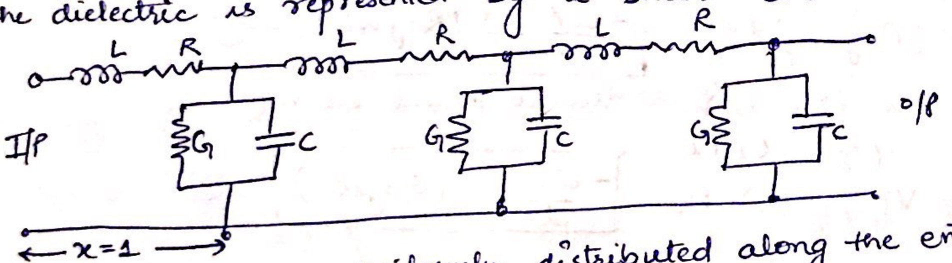


The diameter d, spacing s and length l are dimensional propertie of the line and are important electrically. Electrical properties likes conductivity of wire, dielectric constant, permeability, loss characteristic of any material within the field of line are also important. Tx line can be regarded as a mean of guiding the field so that it is confined near the line rather than spreading in space, spherically. As the wave is confined, it does not suffer inverse square law decrease of power density with distance, unlike waves in free space.

Equivalent Circuit Representation of Transmission line

Just like an ordinary 2-port n/w, a Tx line has inductance(L), capacitance(C) and resistance (R). The essential difference b/w them is that these electrical parameters are uniformly and continuously distributed along the line while ordinary ckt they are lumped.

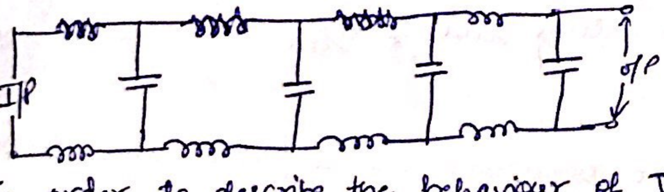
Since each conductor of line has a certain length and diameter, it must have resistance (R) and inductance (L). Since two wires or conductors are close to each other, therefore, there must be capacitance (C) b/w them. Since, the wires are seperated by a medium ~~prop~~ popularly known as dielectric that can not be perfect in its insulation and hence current leaks through it. This leakage of current through the dielectric is represented by a shunt conductance (G).



One unit section of length is represented upto x=1 per unit length (may be per ~~unit~~ meter or per km).

when R, L, G, C are uniformly distributed along the entire length of Tx line. it is said to be uniform Tx line.

At radio frequency (RF), L >> R and capacitive suspension (C) in comparison to shunt conductance (C >> G). This results in a line which is considered to be lossless and thus calculations at RF are simple



In order to describe the behaviour of Tx line, it is helpful to consider two ideal cases which may not be fully attainable in practice but may be closely approximat

1. Infinite length ($l \to \infty$) of Tx line
2. Lossless Tx line (R & G = 0)

The four line parameters e.g. R, L, G and C are referred to as 1° constant of Tx line.

① Resistance (R) :- It is defined as loop resistance per unit length of line. It is sum of resistance of both wires for unit line length. Its unit is Ω/km.

② Inductance (L) : Loop inductance per unit length of line. Sum of inductance of both wires for unit line length and its unit is Henries/km.

③ Conductance (G) : shunt conductance b/w the two conductors per unit length of line and its unit is Mhos/km.

④ Capacitance (C) : shunt capacitance b/w two conductors per unit length of line and its unit is farad/km.

→ These primary constants generally vary with frequency.

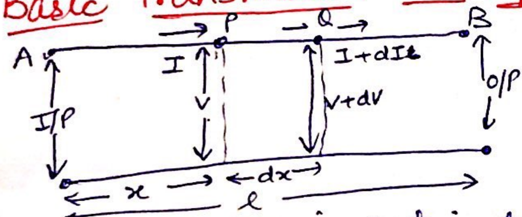
The series impedence (Z) and shunt admittance Y of transmission line per unit length are given by

$$Z = R + j\omega L$$
$$Y = G + j\omega C$$

$\sqrt{\frac{Z}{Y}}$ → characteristic impedance

$\sqrt{ZY}$ → propogation constant

Basic Transmission Line Equation



when dx is small, then
① Voltage b/w conductors can be assumed constant while calculating current &
② current in line may be considered constant while calculating voltages.

Series impedance in section dx → $(R+j\omega L)dx$

shunt admittance in section dx → $(G+j\omega C)\,dx$

∴ voltage drop from P to Q is due to current I flowing through impedance

$$V-(V+dV) = I(R+j\omega L)dx$$
$$-dV = I(R+j\omega L)dx \qquad \boxed{-\frac{dV}{dx} = I(R+j\omega L)} \text{ — ①}$$

|||ly the current difference b/w P, Q is due to shunt admittance

$$\therefore I-(I+dI) = V(G+j\omega C)\,dx$$
$$-dI = V(G+j\omega C)\,dx \qquad \boxed{-\frac{dI}{dx} = V(G+j\omega C)} \text{ — ②}$$

Differentiating ① w.r.t. x

$$-\frac{d^2V}{dx^2} = (R+j\omega L)\frac{dI}{dx}$$

$$\frac{d^2V}{dx^2} = (R+j\omega L)(G+j\omega C)V \quad \rightarrow \quad \boxed{\frac{d^2V}{dx^2} = P^2 V}$$ where $P^2 = (R+j\omega L)(G+j\omega C)$ ↳ propogation constant

|||ly differentiating eq ② w.r.t. x and putting value of dV/dx

$$\boxed{\frac{d^2I}{dx^2} = P^2 I}$$

These are differential eq^n of the wave propogation & is fundamental to the circuit of distributed constant. These equation are standard linear differential eq^n & hence its sol^n in exponential form

$$\boxed{\begin{aligned} V &= Ae^{-Px} + Be^{Px} \\ I &= Ce^{-Px} + De^{+Px} \end{aligned}}$$

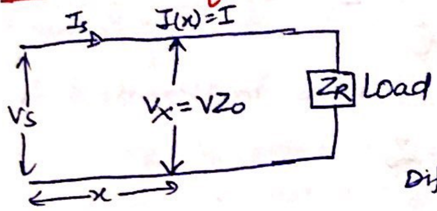
$P^2 = (R+j\omega L)(G+j\omega C)$ or $P = \alpha + i\beta$

so
$$\boxed{\begin{aligned} V &= \underbrace{A e^{-\alpha x} e^{-j\beta x}}_{\text{I term}} + \underbrace{B e^{\alpha x} e^{j\beta x}}_{\text{II term}} \\ I &= C e^{-\alpha x} e^{-j\beta x} + D e^{\alpha x} e^{j\beta x} \end{aligned}}$$

I term $\rightarrow$ There exist a voltage and current component travelling towards the load impedance. These are incident waves which decay exponentially in positive direction of x. i.e. away from source and suffers a phase shift of β radians per unit length of transmission line.

II term $\rightarrow$ A wave similar to incident wave b/w travelling in opposite direction i.e. reflected wave.

Solution of a Transmission Line Terminated with Load Impedance



$V = Ae^{-Px} + Be^{+Px}$ — (1)

$I = Ce^{-Px} + De^{+Px}$ — (2)

Also $-\dfrac{dV}{dx} = I(R+j\omega L)$ — (3)

Diff (1) $\dfrac{dV}{dx} = [Ae^{-Px}(-P) + B^{Px}(P)]$ — (4)

comparing (3) & (4)

$$-[-APe^{-Px} + BPe^{Px}] = I(R+j\omega L)$$

$$P(Ae^{-Px} - Be^{+Px}) = I(R+j\omega L)$$

$$I = \frac{P}{(R+j\omega L)}[Ae^{-Px} - Be^{+Px}] = \frac{\sqrt{(R+j\omega L)(G+j\omega C)}}{(R+j\omega L)}[Ae^{-Px} - Be^{+Px}]$$

$$I = \sqrt{\frac{G+j\omega C}{R+j\omega L}}[Ae^{-Px} - Be^{+Px}]$$

$$I = \frac{1}{Z_0}[Ae^{-Px} - Be^{+Px}] \quad \text{where } Z_0 = \sqrt{\frac{R+j\omega C}{G+j\omega C}} \rightarrow \text{characteristic impedance}$$

We known that $\cosh Px = \dfrac{e^{Px} + e^{-Px}}{2}$ and $\sinh Px = \dfrac{e^{Px} - e^{-Px}}{2}$

$\cosh Px - \sinh Px = e^{-Px}$

$\sinh Px + \cosh Px = e^{Px}$

$$V = Ae^{-Px} + Be^{+Px}$$

$$V = A(\cosh Px - \sinh Px) + B(\cosh Px + \sinh Px)$$

$$\boxed{V = \cosh Px(A+B) - \sinh Px(A-B)} \quad \text{— (5)}$$

$$I = \frac{1}{Z_0}[Ae^{-Px} - Be^{Px}] = \frac{1}{Z_0}[A(\cosh Px - \sinh Px) - B(\cosh Px + \sinh Px)]$$

$$\boxed{I = \frac{1}{Z_0}[\cosh Px(A-B) - \sinh x(A+B)]} \quad \text{— (6)}$$

At the input end, $x=0 \rightarrow$ Voltage $\rightarrow V_s$, current $\rightarrow I_s$, $\cosh Px = 1$ & $\sinh Px = 0$

Applying these conditions in (5) & (6)

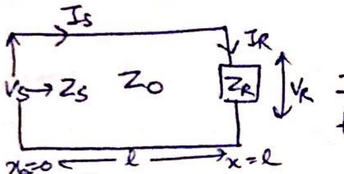
$$V_s = A + B \quad \text{and} \quad I_s = \frac{A-B}{Z_0}$$

Putting these values in equation (5) & (6)

$$\boxed{\begin{aligned} V &= V_s \cosh Px - I_s Z_0 \sinh Px \\ I &= I_s \cosh Px - \left(\frac{V_s}{Z_0}\right)\sinh Px \end{aligned}}$$

Input Impedance of a Transmission line terminated with any load impedance :

$$V = V_S \cosh Px - I_S Z_0 \sinh Px \quad \text{—①}$$
$$I = I_S \cosh Px - \frac{V_S}{Z_0} \sinh Px. \quad \text{—②}$$



If line is terminated at length $x=l$, the voltage across terminated load impedance Z_R is $V = I_R Z_R$ —③

Putting value of V and I from eq ① & ②

$$V_S \cosh Pl - I_S Z_0 \sinh Pl = Z_R\left[I_S \cosh Pl - \frac{V_S}{Z_0}\sinh Pl\right]$$
$$Z_0\left[V_S \cos Pl - I_S Z_0 \sinh Pl\right] = Z_R\left[I_S Z_0 \cosh Pl - V_S \sinh Pl\right]$$
$$V_S\left[Z_0 \cosh Pl + Z_R \sinh Pl\right] = I_S Z_0\left[Z_R \cosh Pl + Z_0 \sinh Pl\right]$$

source impedance $\rightarrow Z_S = \dfrac{V_S}{I_S} = Z_0\left\{\dfrac{[Z_R \cosh Pl + Z_0 \sinh Pl]}{[Z_0 \cosh Pl + Z_R \sinh Pl]}\right\}$ —① $\rightarrow Z_S = Z_0\left[\dfrac{Z_R + Z_0 \tanh Pl}{Z_0 + Z_R \tanh Pl}\right]$

special case → If line is terminated by its characteristic impedance i.e. $Z_R = Z_0$

$$Z_S = Z_0\left[\frac{Z_0 \cosh Pl + Z_0 \sinh Pl}{Z_0 \cosh Pl + Z_0 \sinh Pl}\right] \qquad Z_S = Z_0$$

Input Impedance of a Lossless R.f. Transmission line or low loss transmission line :

For a lossless Tx line $R=0, G=0$ and $\alpha = \dfrac{R}{2Z_0}$ = attenuation constant $= 0$ $\qquad \beta = \dfrac{2\pi}{\lambda}$

∴ Propogation constant $P = \alpha + i\beta \approx i\beta$

Putting this in ①

$$Z_S = Z_0\left[\frac{Z_R \cosh(j\beta)l + Z_0 \sinh(j\beta)l}{Z_0 \cosh(j\beta)l + Z_R \sinh(j\beta)l}\right]$$

$\cosh j\beta l = \cos \beta l$
$\sinh j\beta l = j \sin \beta l$

$$Z_S = Z_0\left[\frac{Z_R \cos\beta l + jZ_0 \sin\beta l}{Z_0 \cos\beta l + jZ_R \sin\beta l}\right] \quad \text{or} \quad Z_S = Z_0\left[\frac{Z_R + jZ_0 \tan\beta l}{Z_0 + jZ_R \tan\beta l}\right]$$

Infinite Transmission Line

If a line of infinite length is considered, then all the power fed into it will be absorbed bcz as we move away from source terminal, I and V will decrease along the line and become zero at an infinite distance.

At $x=0$, $V=V_S$ $\qquad e^0 = \frac{1}{e^0} = \frac{1}{1} = 1$

$V = Ae^{-Px} + Be^{Px}$ $\longrightarrow V_S = Ae^{-P\cdot 0} + Be^{+P\cdot 0} \Rightarrow \boxed{V_S = A+B}$ —①

$0 = Ae^{-\infty} + Be^{\infty} \Rightarrow 0 + Be^{\infty} = 0 \Rightarrow \boxed{B=0}$ —② $\qquad e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$

At $x=\infty$ $V=V_0$

Putting ② in ① $\qquad V_S = A \Rightarrow \boxed{V = V_S e^{-Px}}$

Diff wrt x

$$\frac{dV}{dx} = -PV_S e^{-Px} \qquad -\frac{dV}{dx} = I(R+j\omega L)$$

$$I(R+j\omega L) = PV_S e^{-Px}$$

$$I = \frac{PV_S}{R+j\omega C}\cdot e^{-Px} = \frac{\sqrt{(R+j\omega L)(G+j\omega C)}}{(R+j\omega L)} V_S \cdot e^{-Px}$$

$$I = \sqrt{\frac{G+j\omega C}{R+j\omega L}}\cdot V_S \cdot e^{-Px}$$

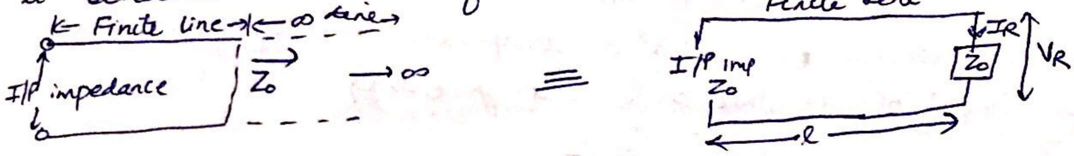
$$\boxed{I = \frac{V_S e^{-Px}}{Z_0}}$$

$$\frac{V}{I} = \frac{V_S e^{-Px}}{\frac{V_S e^{-Px}}{Z_0}} \qquad \boxed{\frac{V}{I} = Z_0}$$

∴ I/P impedance of an infinite line is called characteristic impedance

An Infinite Line is Equivalent to a finite line terminated in its Characteristic Impedance

When a finite length of line is joined with an infinite line, their input impedance is same as that of infinite line, because they together make an infinite length. The infinite ~~length~~ line presents an impedance Z_0 at its input (Z_0 = characteristic impedance). Therefore, it can be concluded that a finite line has an I/P impedance Z_0 when it is terminated in Z_0 or that when a finite line is terminated by Z_0 it behaves as an infinite line.



$Z_0 = \frac{V_R}{I_R}$ → for finite line

Putting $x = l$, $V = V_R$ and $I = I_R$ in general eqⁿ

$$V_R = V_S \cosh Pl - I_S Z_0 \sinh Pl$$

$$I_R = V_S \cosh Pl - \frac{V_S}{Z_0} \sinh Pl$$

$$\frac{V_R}{I_R} = \frac{V_S \cosh Pl - I_S Z_0 \sinh Pl}{V_S \cosh Pl - \frac{V_S}{Z_0} \sinh Pl} = Z_0 \qquad \because Z_0 = \frac{V_R}{I_R}$$

$$V_S \cosh Pl - I_S Z_0 \sinh Pl = Z_0\left[V_S \cosh Pl - \frac{V_S}{Z_0} \sinh Pl\right]$$

$$V_S[\cosh Pl + \sinh Pl] = Z_0 I_S[\cosh Pl + \sinh Pl]$$

$Z_0 = \frac{V_S}{I_S}$ → I/P impedance of line

$\therefore \boxed{Z_S = \frac{V_S}{I_S} = \frac{V_R}{I_R} = Z_0}$ ⇒ A finite Tx line when terminated by its characteristic impedance Z_0, it is equivalent to an infinite line.

Characteristic Impedance (also known as surge impedance) (ET-2017 along with eqⁿ)

It is defined as:

① As the steady state vector ratio of the voltage to the current at the input of an infinite line OR

② As the impedance measured at the I/P of this line when its length is finite or impedance into an infinite length of line.

Proof:
$$-\frac{dV}{dx} = I(R+j\omega L) \qquad V = V_S e^{-Px} \text{ and } I = I_S e^{-Px}$$

$$\therefore -\frac{d}{dx}\left[V_S e^{-Px}\right] = I_S e^{-Px}[R+j\omega L]$$

$$V_S e^{-Px} \cdot P = I_S e^{-Px}[R+j\omega L]$$

$$\frac{V_S}{I_S} = \frac{1}{P}[R+j\omega L] = \frac{1}{\sqrt{(R+j\omega L)(G+j\omega C)}} \cdot [R+j\omega L]$$

→ characteristic impedance of a line can be measured at its I/P by terminating the line at far end with an impedance equal to characteristic impedance.

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{Z}{Y}}$$

Z → series impedance
Y → shunt admittance

Characteristic impedance does not depend on length of line or character of terminal load.

→ characteristic impedance depends only on the characteristic of the line per unit length.

Characteristic impedance for two basic types of Tx line is

$$Z_0 = \frac{276}{\sqrt{K}} \log_{10} \frac{S}{r} \, \Omega \rightarrow \text{||el wire}$$

$$Z_0 = \frac{138}{\sqrt{K}} \log_{10} \frac{D}{d} \, \Omega \rightarrow \text{coaxial}$$

$$Z_0 = 276 \log_{10} \frac{S}{r} \, \Omega \leftarrow \text{for dielectric parallel wire line as } \sqrt{K} = 1 \text{ for Air}$$

K → dielectric constant
D & d → inner & outer radius of conductor
S → Centre to centre spacing b/w two conductors
r → radius of each conductor.

At radio (higher) frequencies, the resistive component of the equivalent circuit becomes negligible i.e. $j\omega L >> R$ and $j\omega C >> G$

$$\therefore Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} \, \Omega$$

Propogation Constant (Attenuation and Phase Constant)

→ Propogation constant reveals the nature in which wave is propogated along the line i.e. how V and I vary with distance x.

$$P = \sqrt{(R + j\omega L)(G + j\omega C)} \quad \text{or} \quad P = \sqrt{ZY}$$

<u>Defination</u> :- The natural logarithm of the vector ratio of the steady state voltages or currents entering & leaving the structure.

$$\text{i.e. } P = \log_e \frac{I_S}{I_R} \quad \text{or} \quad P = \log_e \frac{V_S}{V_R}$$

Propogation constant × length of tx line = Electrical length of line

$$P = \alpha + i\beta = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{j\omega L \cdot j\omega C \left(1 + \frac{R}{j\omega L}\right)\left(1 + \frac{G}{j\omega C}\right)}$$

$$P = \sqrt{j^2\omega^2 LC} \sqrt{\left(1 + \frac{R}{j\omega L}\right)} \sqrt{1 + \frac{G}{j\omega C}}$$

$$= j\omega\sqrt{LC} \sqrt{1 + \frac{R}{j\omega L}} \sqrt{1 + \frac{G}{j\omega C}}$$

Since $\omega L >> R$ and $\omega C >> G$ $\quad \frac{R}{\omega L} << 1$ and $\frac{G}{\omega C} << 1$

Using binomial expansion

$$P = j\omega\sqrt{LC}\left[\left(1 + \frac{1}{2}\frac{R}{j\omega L} + \cdots\right)\left(1 + \frac{1}{2}\frac{G}{j\omega C} + \cdots\right)\right]$$

$$P = j\omega\sqrt{LC}\left[1 + \frac{1}{2}\frac{G}{j\omega C} + \frac{R}{2j\omega L} + \cdots\right]$$

$$= j\omega\sqrt{LC} + \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}$$

$$\boxed{\alpha = \frac{R}{2}\sqrt{\frac{L}{C}} + \frac{G}{2}\sqrt{\frac{L}{C}}} \qquad \boxed{\beta = \omega\sqrt{LC}}$$

Also $\boxed{v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}}$

α → Attenuation constant (Neper)
↳ reduction of voltage & current along the line

β → phase constant → variation in phase position of voltage & current along the line

$$\boxed{\begin{array}{l} 1 \text{ neper} = 8.686 \text{ dB} \\ 1 \text{ radian} = 57.3^\circ \end{array}}$$

Derivation of Expression for α and β in term of Primary constant (R, L, G, C)

(ET 2018, 6.5M)

The propogation constant (P), Attenuation constant (α), phase constant (β) and characteristic impedance (Z_0) are called secondary coefficients or secondary constant of transmission line.

Since $= P = \sqrt{(R+j\omega L)(G+j\omega C)} = \alpha + j\beta$

$$P^2 = (R+j\omega L)(G+j\omega C) = \alpha^2 - \beta^2 + 2j\alpha\beta$$

$$RG + j\omega CR + j\omega LG - \omega^2 LC = \alpha^2 - \beta^2 + 2j\alpha\beta$$

$$\alpha^2 - \beta^2 + 2j\alpha\beta = RG - \omega^2 LC + j\omega(LG + RC)$$

$\therefore$ $\boxed{\alpha^2 - \beta^2 = RG - \omega^2 LC}$ — ① and $2\alpha\beta = \omega(LG + RC)$

$$|P| = \sqrt{\alpha^2+\beta^2} = \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)} \rightarrow \alpha^2+\beta^2 = \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)} \text{ — ②}$$

Adding ① & ②

$$2\alpha^2 = RG - \omega^2 LC + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$$

$$\boxed{\alpha = \pm\sqrt{\tfrac{1}{2}\left[(RG-\omega^2LC) + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}\right]}}$$

Subtracting

$$\boxed{\beta = \pm\sqrt{\tfrac{1}{2}(\omega^2LC - RG) + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}}}$$

In practice, $R >> \omega L$ and $\omega C >> G$

So, $P = \sqrt{(j\omega C)\times R} = \alpha + j\beta$

$$= \sqrt{\omega CR}\angle 45^\circ$$

$$= \sqrt{\omega CR}(\cos 45^\circ + j\sin 45^\circ) = \frac{1}{\sqrt{2}}\sqrt{\omega RC} + \frac{1}{\sqrt{2}}j\sqrt{\omega CR}$$

$$\therefore \boxed{|\alpha| = |\beta| = \sqrt{\frac{\omega CR}{2}}}$$

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{R}{j\omega C}\times\frac{j}{j}} = \sqrt{\frac{-R}{j\omega C}} = \sqrt{\frac{R}{\omega C}\times(-1)} \qquad \boxed{Z_0 = \sqrt{\frac{R}{\omega C}}\angle -45^\circ}$$

Parameters of Transmission Line

① **Wavelength (λ)**: It is the distance in which the phase change of 2π radians is effected by a wave travelling along the line $\boxed{\lambda = \frac{2\pi}{\beta}}$

② **Phase Velocity (V_p)**:- Velocity with which a signal of frequency propogates along the line at a particular frequency (f). Measured in km/s. It is the velocity of propogation along the line based on observation of phase change along the line hence called phase velocity

$$\lambda = V_p \times t \rightarrow V_p = \lambda f \qquad \boxed{V_p = \frac{\omega}{\beta}}$$

③ **Group Velocity (V_g)**:- When diff. freq. travel with diff. velocities, then line or medium is said dispersive, then velocity is called group velocity (V_g)

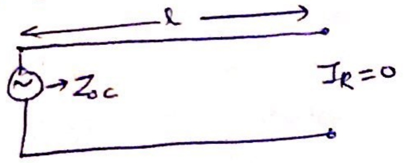
$V_g < V_p$ $\qquad V_g = \frac{d\omega}{d\beta}$

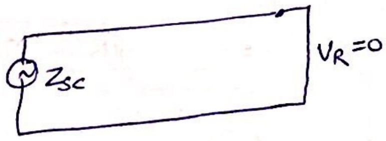
Open and Short Circuit Lines

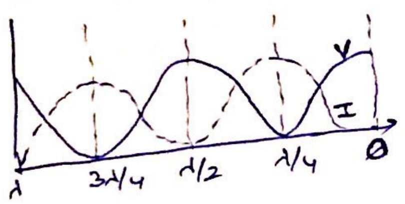
In any transmission line the terminating load establishes the current & voltage relations. Hence three basic type (important) terminating load are:-

① when terminating end is open circuited → Load end is open
② when terminating end is short circuited → end is shorted by metal strip
③ when terminating load is equal to characteristic impedance.

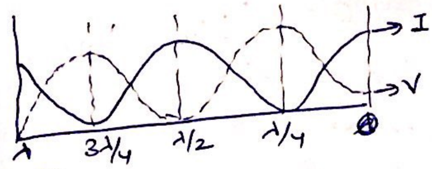
Open ckt and short ckt Tx line of suitable length can be arranged to act as series resonant or parallel resonant or desired value of Reactance (L or C). At higher frequencies length of such Tx lines is small and they are used as reactance. These short length of Tx line is called STUBS. Short ckted stubs are preferred then open ckted stubs (at ↑ freq) due to radiation losses at end.

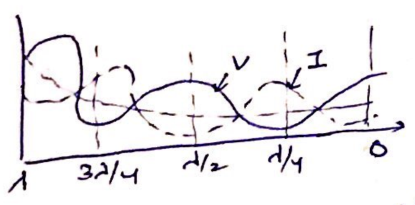




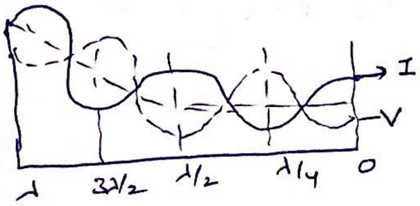


lossless line





lossy line



Point of voltage max → antinodes
" " min → nodes

Since in case of open ckt and close ckt complete reflection occurs and thus there is complete cancellation of voltage min is zero. In other words voltage maxima (antinodes) and voltage minima (nodes) remain motionless and so standing wave is said to exist on the line

① **Open ckted Line** :- Voltage → Max & current ≈ 0 ⇒ Z = ∞
At λ/4 from the load end incident wave is 90° earlier and reflected wave is 90° later than what they are at load end, ∴ phase difference of 180° resulting in a minima at λ/4 point. ∴ Standing wave pattern at every λ/2 length.

② **Short ckted lines** :- current Max, voltage ≈ 0, Z=0. Here the standing wave has minimum voltage or node at short ckted end the voltage minima are spaced at every λ/2 apart from the load end.

③ **Line terminated with Z_0** :- This line is said to be correctly terminated and all power sent from source is absorbed in Load, hence there is no power loss or reflection

$V_R = V_s \cosh Pl - I_s Z_0 \sinh Pl$

$I_R = I_s \cosh Pl - \frac{V_s}{Z_0} \sinh Pl$

when OC $I_R = 0$ $\rightarrow 0 = I_s \cosh Pl - \frac{V_s}{Z_0} \sinh Pl$ $\quad Z_{oc} = \frac{V_s}{I_s} = Z_0 \frac{\cosh Pl}{\sinh Pl} = Z_0 \coth Pl$

when SC $V_R = 0$ / $0 = V_s \cosh Pl - I_s Z_0 \sinh Pl \Rightarrow Z_{sc} = \frac{V_s}{I_s} = Z_0 \frac{\sinh Pl}{\cosh Pl} = Z_0 \tanh Pl$

Multiplying $Z_{sc} \times Z_{oc} = Z_0 \coth Pl \cdot Z_0 \tanh Pl = Z_0^2$

$\boxed{Z_0 = \sqrt{Z_{sc} \cdot Z_{oc}}}$ as $\frac{Z_{sc}}{Z_{oc}} = \frac{Z_0 \tanh Pl}{Z_0 \coth Pl}$ $\quad \boxed{\tanh Pl = \sqrt{\frac{Z_{sc}}{Z_{oc}}}}$

So, by measuring Z_{sc} and Z_{oc} $\rightarrow$ Z_0 can be calculated.

Z_{sc} & $Z_{oc} \rightarrow Z_0$ & $P \rightarrow R, L, G, C$

Open and Short Circuited Lossless Transmission Lines

$V_R = V_s \cosh Pl - Z_0 I_s \sinh Pl$

$I_R = I_s \cosh Pl - \frac{V_s}{Z_0} \sinh Pl$

For lossless line $\alpha = 0$ $\quad \therefore P = \alpha + j\beta = j\beta$

$V_R = V_s \cosh j\beta l - I_s Z_0 \sinh j\beta l$

$= V_s \cos\beta l - j I_s Z_0 \sin\beta l$ $\qquad \cosh j\beta = \cos\beta$, $\sinh j\beta = j \sin\beta$

|||ly $I_R = I_s \cos\beta l - j \frac{V_s}{Z_0} \sin\beta l$

OC $(z = \infty)$

$0 = I_s \cos\beta l - j \frac{V_s}{Z_0} \sin\beta l$

$I_s \cos\beta l = j \frac{V_s}{Z_0} \sin\beta l$

$Z_{oc} = \frac{V_s}{I_s} = \frac{Z_0 \cos\beta l}{j \sin\beta l} \times \frac{j}{j}$

$\boxed{Z_{oc} = -Z_0 j \cot\beta l}$ ← inductive when $\cot\beta l$ is −ve, capacitive when $\cot\beta l$ is +ve

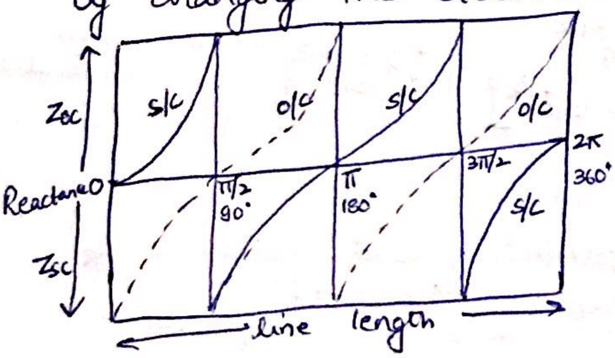
SC $(z = 0)$

$0 = V_s \cos\beta l - j I_s Z_0 \sin\beta l$

$\frac{V_s}{I_s} = j Z_0 \frac{\sin\beta l}{\cos\beta l}$

$\boxed{Z_{sc} = j Z_0 \tan\beta l}$ ← inductive when $\tan\beta l$ is +ve, capacitive when $\tan\beta l$ is −ve

I/P impedance of SC and OC lossless Tx line is pure reactace as viewed from the sending end. The value of reactance can be changed by changing the electrical length βl stubs.



Open ckt and short ckt. lines behave as resonant circuit. This happens when length of line is $\lambda/4$ (multiples of $\lambda/4$)

$\beta l = n\lambda/4$

then $\tan\beta l = \infty$ and $\cot\beta l = 0$

$\therefore Z_{oc} = -j \cot\left(\frac{2\pi}{\lambda} \cdot \frac{n\lambda}{4}\right) = -j \cot\frac{\pi}{2} = 0$

$\boxed{Z_{oc} = 0}$

$Z_{sc} = j \tan\left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{4}\right) = j \tan\frac{\pi}{2} = \infty$

$\boxed{Z_c = \infty}$

⇒ $\lambda/4$ SC line present ∞ Impedance at input terminal just like a parallel resonance (LC) circuit and a $\lambda/4$ OC line present zero I/P impedance like a series LC ckt.

⇒ Transmission line act as a resonant ckt and is used in many applications at high frequencies in antenna design and other radio ckt.

Reflection

The phenomenon of setting up of a reflected wave at the [cut off] due to improper termination or due to impedance irregularity in a line is known as reflection. The reflection occurs in all cases of travelling wave where discontinuity of medium is involved.

→ Although reflection is normally undesirable on the transmission line Reflection wave may appear as echo at sending end provided attenuation is not large. It may further be reflected from the sending end and thus energy is oscillating back and forth on the line until die out of line losses.

→ Reflection is extreme when the line is open or short circuited and zero when $Z_R = Z_0$.

Open Circuit Condition:- When incident wave approaches the end, $\vec{H}$ collapes because current is becoming zero. This collapsing magnetic field produces an electric field which is added to the existing electric field and hence voltage at open circuited end is increased. This additional voltage gives rise to a wave which travels back to Sending end. Since no energy is absorbed at open end and so the reflected wave has same magnitude and phase what the incident wave has at the termination.

Short Circuit Condition:- $\vec{E}$ field collapses on the shorting ~~stip~~ strip, and a the charge on the line is reversed at the short circuit and a reflected wave is set up again. At the front of reflected wave a current doubling is said to occur when voltage is zero.

Reflection Coefficients

Reflection Coefficients:- It is the ratio of reflected voltage (or current) to the incident voltage (or current). It is denoted by K or r or ρ.

It is a vector quantity having magnitude and direction as well

$$\frac{V_r}{V_i} = K \qquad \text{also} \qquad -K = \frac{I_r}{I_i}$$

Fundamental Tx line eq^n

$$V = Ae^{-Px} + Be^{+Px}$$
$$I = \frac{1}{Z_0}\left[Ae^{-Px} - Be^{+Px}\right]$$

{eq ① & ⑤ in derivation of sol^n of Tx line terminated with load impedance}

I term → incident wave voltage

& II term → Reflected wave voltage

Let y be a distance measured from the terminating load impedance Z_R then $x = -y$

$$V = Ae^{Py} + Be^{-Py}$$
$$I = \frac{1}{Z_0}\left[Ae^{Py} - Be^{-Py}\right]$$

Here also I term → Incident & II term → Reflected

when $y = 0$ $V = V_R$ and $I = I_R$ $\quad V_R = Ae^{0} + Be^{-0}$ $\quad V_R = A + B$

$$I_R = \frac{A-B}{Z_0} \qquad Z_0 I_R = A - B$$

$$A = \frac{V_R + Z_0 I_R}{2} \qquad \text{and} \qquad B = \frac{V_R - Z_0 I_R}{2}$$

$$K = \frac{V_r}{V_i} = \frac{Be^{-Py}}{Ae^{Py}} = \frac{B}{A}e^{-2Py} = \frac{\frac{V_R - Z_0 I_R}{2}}{\frac{V_R + Z_0 I_R}{2}} e^{-2P\cdot 0} = \frac{V_R - Z_0 I_R}{V_R + Z_0 I_R} = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$\boxed{K = \frac{Z_R - Z_0}{Z_R + Z_0}}$$

Input Impedance in term of Reflection Coefficient

I/P impedance $Z_S = Z_0\left[\frac{Z_R \cosh Pl + Z_0 \sinh Pl}{Z_0 \cosh Pl + Z_R \sinh Pl}\right]$

$\cosh Pl = \frac{e^{Pl} + e^{-Pl}}{2}$

$\sinh Pl = \frac{e^{Pl} - e^{-Pl}}{2}$

Changing $\cosh Pl$ & $\sinh Pl$ in exponential terms

$$Z_S = Z_0\left[\frac{Z_R\left(\frac{e^{PL} + e^{-PL}}{2}\right) + Z_0\left(\frac{e^{PL} - e^{-PL}}{2}\right)}{Z_0\left(\frac{e^{+Pl} + e^{-Pl}}{2}\right) + Z_R\left(\frac{e^{PL} - e^{-PL}}{2}\right)}\right]$$

$$Z_S = Z_0\left[\frac{\frac{e^{PL}}{2}(Z_R + Z_0) + \frac{e^{-PL}}{2}(Z_R - Z_0)}{\frac{e^{PL}}{2}[Z_0 + Z_R] - \frac{e^{-PL}}{2}[Z_R - Z_0]}\right] \Rightarrow Z_S = Z_0\left[\frac{1 + e^{-2PL}\left(\frac{Z_R - Z_0}{Z_R + Z_0}\right)}{1 - e^{-2PL}\left(\frac{Z_R - Z_0}{Z_R + Z_0}\right)}\right]$$

$$\boxed{Z_S = Z_0\frac{(1 + Ke^{-2Pl})}{(1 - Ke^{-2Pl})}}$$

Standing Wave Ratio

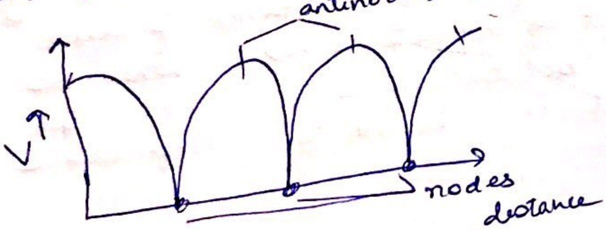
When reflection occurs in an incorrectly terminated line, the incident & the reflected wave give rise to a interference phenomenon & thus standing waves are formed, with definite maxima and minima of current & voltage.

$S = SWR$ = ratio of maximum to minimum current or voltage on a line having standing wave.

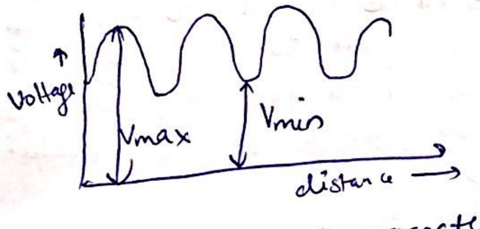
while dealing it voltage it is also abbrivated as VSWR.

$$VSWR = \left|\frac{V_{max}}{V_{min}}\right| \quad \text{or} \quad S = \left|\frac{V_{max}}{V_{min}}\right| = \left|\frac{I_{max}}{I_{min}}\right|$$

→ The SWR is a mismatch b/w load and transmission line.



SWR on a ~~incorrectly terminated line~~ OC & SC line



SWR on a incorrectly terminated line

$V_{max} = |V_i| + |V_r|$

$V_{min} = |V_i| - |V_r|$

$$\frac{V_{max}}{V_{min}} = \frac{|V_i| + |V_r|}{|V_i| - |V_r|} = \frac{1 + \left|\frac{V_r}{V_i}\right|}{1 - \left|\frac{V_r}{V_i}\right|}$$

$$\boxed{VSWR = \frac{1 + |K|}{1 - |K|}}$$

Applying C&D $\frac{VSWR}{1} = \frac{1+K}{1-|K|} \Rightarrow \frac{VSWR-1}{VSWR+1} = \frac{1+|K|-1+|K|}{1+|K|+1-|K|} = \frac{2|K|}{2}$

Relation b/w R & VSWR (2017, 2018)

$$\boxed{|K| = \frac{VSWR-1}{VSWR+1}}$$

Location of Voltage Maxima and Minima

$V = Ae^{+Px} + Be^{-Py}$

For a lossless Tx line $\alpha = 0 \Rightarrow P = j\beta$

$V_x = Ae^{j\beta y} + Be^{-j\beta y}$ where y = distance measured from load end Z_R

Reflection coefficient $K = \frac{V_r}{V_i} = \frac{B \cdot e^{-j\beta y}}{Ae^{j\beta y}} = \frac{B}{A}\frac{e^{-j\beta \cdot 0}}{e^{j\beta 0}} = \frac{B}{A}$ at receiving end $y=0$

$$\boxed{K = \frac{B}{A}}$$

Reflection coefficient in polar form $K = K\angle\theta = |K|e^{j\theta} = \frac{B}{A}$ or $\boxed{B = A|K|e^{j\theta}}$

$V_x = Ae^{j\beta y} + A|K|e^{j\theta} \cdot e^{-j\beta y}$

$= Ae^{j\beta y}\left[1 + |K|e^{j\theta}\, e^{-2j\beta y}\right]$

$$\boxed{V_x = Ae^{j\beta y}\left[1 + |K|e^{-j(2\beta y - \theta)}\right]}$$

Taking magnitude only $|V_x| = |A|\left[1+|K|e^{-j(2\beta y-\theta)}\right]$

So, voltage have maximum amplitudes when the two components are in phase

i.e. $y = Y_{max}$ when $2\beta y - \theta = 2n\pi \rightarrow V_{max} = |A|[1+|K|]$

& $y = Y_{min}$ when $2\beta y - \theta = (2n+1)\pi \rightarrow V_{min} = |A|[1-|K|]$

$VSWR = \left|\frac{V_{max}}{V_{min}}\right| = \frac{A[1+K]}{A[1-K]} = \frac{1+|K|}{1-|K|}$

Impedance Matching

For the maximum transfer of power b/w source & load, the resistance of load should be equal to that of source and the reactance of load should be equal to that of source but of opposite sign i.e. when the source is inductive, the load should be capacitive and vice-versa. When this condition is achieved it is referred to as impedance matching

Principle of Impedance Matching

$$\text{Radio Transmitter} \longrightarrow \underbrace{\text{Tx line} \longrightarrow \text{Antenna}}_{\text{considered as a single unit}}$$

$Z_g = R_g + jX_g$ $\qquad Z_s = R_s - jX_s$

For max power Tx $\rightarrow \boxed{R_g = R_s}$ and $\boxed{X_g = -X_s}$

Z_s depends on Z_R, Z_0 and length l.

Practically l is determined by the necessary spacing b/w the Tx and the receiver antenna & hence can't be independent adjustable. Range of Z_0 is also practically limited

To minimise the SWR, $\boxed{Z_R = Z_0}$ or $\boxed{Z_S = Z_0}$

So Radio Tx → Impedance Transformer → Line ~~line~~ I/P line O/P → Impedance Transformer → Antenna ~~Tran~~ Terminal

Impedance Transformer (after Radio Tx) ↓ Transform the Z_0 into a value for Max power Tx

Impedance Transformer (before Antenna) ↓ Transform the antenna impedance (Z_0) into characteristic impedance Z_0.

These impedance transformers of impedance may consist of coupled coil of wire or of coil and condenser ckt at low frequencies. At higher frequencies (in MHz) these impedance transformer usually consists of section of Tx line in various arrangments.

One of such possible arrangement is <u>STUB MATCHING</u>

<u>Stub Matching</u>

Connection of sections of open and short circuited line known as stub / tuning stub in shunt with the main line at a certain point (or points) to effect the matching ~~this~~ is known as stub matching,

<u>Advantages of Stub Matching</u>

① Length (l) and characteristic impedance (Z_0) remains unchanged.

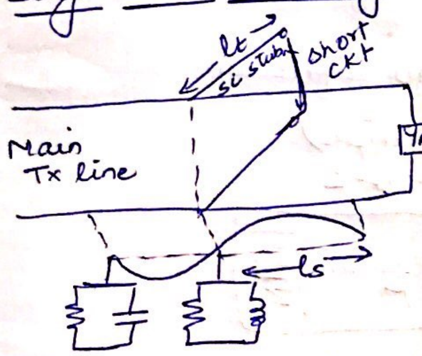
② Mechanically, it is possible to add adjustable susceptance in shunt with the line.

Stub matching is of two Types < Single stub Matching / Double stub Matching

Impedance = Resistance + Reactance
↓ Real, ↓ Imaginary
↕
Admittance = Conductance + Susceptance
↓ Real, ↓ Imaginary

$Y = G + jB$

<u>Single Stub Matching</u>



<u>Principal</u> → A SC section of a line whose open end is connected to the line at a particular distance from the load end, where the I/P conductance at this point is equal to characteristic conductance of the line.

→ The stub length is adjusted to provide a susceptance which is equal in value but opposite in sign to the I/P susceptance of the main line at that point, so that total susceptance at the point of attachement is zero. ~~The~~

→ The combination of stub and line will thus present a conductance which is equal to the Z_0 of line. So, main length of HFTL will be matched.

→ Stub should be located as near to the load as possible.

To find length of stub (l_s) and length of SC stub (l_t)

$$Z_s = Z_0 \frac{Z_R + Z_0 \tanh Pl}{Z_0 + Z_R \tanh Pl} \qquad Y_0 = \frac{1}{Z_0},\ Y_R = \frac{1}{Z_R} \ \& \ Y_s = \frac{1}{Z_s}$$

Since the stub is connected in parallel with main line so it is better to do calculation in admittance (Y) rather than impedance so that it can directly be added up.

$$Y_s = \frac{1}{Z_s} = \frac{Z_0 + Z_R \tanh Pl}{Z_0(Z_R + Z_0 \tanh Pl)} = \frac{Y_0\left(\frac{1}{Y_0} + \frac{1}{Y_R}\tanh Pl\right)}{\left(\frac{1}{Y_R} + \frac{1}{Y_0}\tanh Pl\right)} = \frac{Y_0(Y_R + Y_0 \tanh Pl)}{(Y_0 + Y_R \tanh Pl)}$$

At high freq $\alpha = 0 \Rightarrow P = j\beta$

$$Y_s = Y_0\left(\frac{Y_R + jY_0 \tanh \beta l}{Y_0 + jY_R \tanh \beta l}\right) \qquad y_s = \frac{Y_s}{Y_0} = \text{Normalized I/P admittance}$$

$$\frac{Y_s}{Y_0} = \left(\frac{\frac{Y_R}{Y_0} + j\tan\beta l}{1 + j\frac{Y_R}{Y_0}\tan\beta l}\right) \rightarrow y_s = \frac{y_r + j\tan\beta l}{1 + j y_r \tan\beta l} \qquad y_r = \frac{Y_R}{Y_0} = \text{Normalized load admittance}$$

$$y_s = \frac{y_r + j\tan\beta l}{1 + j y_r \tan\beta l} \times \frac{1 - j y_r \tan\beta l}{1 - j y_r \tan\beta l} = \frac{y_r - j y_r^2 \tan\beta l + j\tan\beta l - j^2 y_r \tan^2\beta l}{1 + (y_r \tan\beta l)^2}$$

$$y_s = G_s + jB_s = \frac{y_r(1 + \tan^2\beta l) + j\tan\beta l(1 - y_r^2)}{[1 + (y_r\tan\beta l)^2]}$$

$$\therefore G_s = \frac{y_r(1 + \tan^2\beta l)}{1 + (y_r \tan\beta l)^2} \qquad B_s = \frac{\tan\beta l(1 - y_r^2)}{1 + (y_r\tan\beta l)^2}$$

$y_s = G_s + jB_s = 1 = 1 + j0 \Rightarrow y_s = G_s = 1$ and $B_s = 0$ for No Reflection

Stub has to be connected where there is no reflection

$$G_s = 1 = \frac{y_r(1 + \tan^2\beta l_s)}{1 + (y_r \tan\beta l_s)^2} \qquad 1 + y_r^2 \tan^2\beta l_s = y_r(1 + \tan^2\beta l_s)$$

$$y_r \tan^2\beta l_s (y_r - 1) = y_r - 1$$

$$\tan^2\beta l_s = \frac{1}{y_r} = \frac{1}{\frac{Y_R}{Y_0}} = \qquad \tan\beta l_s = \frac{1}{\sqrt{\frac{Y_0}{Y_R}}} \qquad \boxed{l_s = \frac{1}{\beta}\tan^{-1}\sqrt{\frac{Z_R}{Z_0}}}$$

Susceptance at the point of attachement of stub

$$b_s = \frac{B_s}{Y_0} = \frac{\tan\beta l_s(1 - y_r^2)}{1 + (y_r \tan\beta l_s)^2} = \frac{\sqrt{\frac{Y_R}{Y_0}}\left(1 - \frac{Y_R^2}{Y_0^2}\right)}{1 + \frac{Y_R^2}{Y_0^2} * \frac{Y_0}{Y_R}} = \frac{\sqrt{\frac{Y_0}{Y_R}}\,(Y_0^2 - Y_R^2)}{\left(1 + \frac{Y_R}{Y_0}\right)Y_0^2}$$

$$\boxed{B_s = \sqrt{\left(\frac{Y_0}{Y_R}\right)}\,(Y_0 - Y_R)}$$

The fundamental eqn is $V_R = V_s \cos\beta l - jZ_0 I_s \sin\beta l$

for SC stub $V_R = 0$ $\qquad Z_t = \frac{V_s}{I_s} = jZ_0 \tan\beta l_t$

$$\boxed{Y_t = \frac{1}{Z_t} = -jY_0 \cot\beta l_t}$$

$Y_t = G_t + jB_t$ $\quad G_t = 0 \quad B_t = -Y_0 \cot\beta l_t$

At point of attachment, sum of line susceptance & stub susceptance must be zero $\quad B_S + B_t = 0$

$$\text{or } \sqrt{\frac{Y_0}{Y_R}}(Y_0 - Y_R) + \{-Y_0 \cot\beta l_t\} = 0$$

$$\cot\beta l_t = \frac{(Y_0 - Y_R)}{Y_0}\sqrt{\frac{Y_0}{Y_R}} \qquad \tan\beta l_t = \frac{\sqrt{Y_0 Y_R}}{Y_0 - Y_R}$$

$$\boxed{l_t = \frac{\lambda}{2\pi}\tan^{-1}\frac{\sqrt{Y_0 Y_R}}{Y_0 - Y_R}}$$

→ The susceptance of line is cancelled & line appears to be terminated by pure R_0.

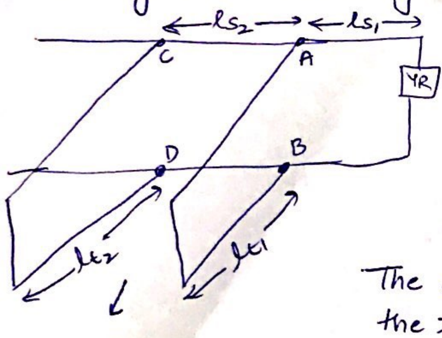
Advantages of Single stub Matching:

1. Less Power Radiation
2. Effective length variation is possible by a shorting bar, therefore, a short circuited stub is invariably used.

Demerits:

1. The range of terminating impedances which can be tranferred is limited. If terminating impedance changes, it becomes essential to adjust the position and length of stub as well.
2. If is useful for a fixed frequency only bcz as frequency varies, the position of stub has to be varied.

Doubling Stub Matching



Here two short ckt stubs are placed at two fixed points usually $\lambda/4$ apart. Although position is fixed but their length are independently adjustable. The double stub matching provides wide range of impedance matching.

The spacing b/w two stubs is 0.375λ or $\lambda/4$.

The spacing of $\lambda/4$ is optimum bcz spacing of $\lambda/2$ places the two stubs in parallel which is similar to the case when two stubs are placed very close together.

usually employed in coaxial MW lines

$$(Y_s)_{AB} = \frac{Y_r + j\tan\beta l_{s_1}}{1 + jY_r\tan\beta l_{s_1}} \qquad (Y_s)_{AB} = \frac{Y_r + j\tan\beta l_{s_1}}{Y_r + j\tan\beta l_{s_1}} \times \frac{1 - jY_r\tan\beta l_{s_1}}{1 - jY_r\tan\beta l_{s_1}}$$

$$(Y_s)_{AB} = \underbrace{\frac{Y_r\sec^2\beta l_{s_1}}{1 + Y_r^2\tan^2\beta l_{s_1}}}_{(G_s)_{AB}} + j\underbrace{\frac{(1 - Y_r^2)\tan\beta l_{s_1}}{1 + Y_r^2\tan^2\beta l_{s_1}}}_{(B_s)_{AB}}$$

When a stub having susceptance B_1 is added to this point AB, the new admittance will be $(Y_s)_{CD} = (G_s)_{CD=AB} + j(B_s)_{CD}$ ($\because$ only B changes by addition of stub & not G)

$\therefore (Y_s)_{CD}$ should be such that admittance Y at CD is equal to $1 + jB_2$

The stub length of CD is adjusted such that the new value of $(Y_S)_{CD}$ is unity to effect the matching.

i.e. $y_S = 1$ $\quad y_S = \frac{Y_S}{G_0} = 1$ $\quad$ or $\boxed{Y_S = G_0}$

Hence the point CD should be at a location on line having a per unit admittance of $y_S = \frac{Y_S}{G_0} = \boxed{1 \pm j\beta_2}$

→ The stub AB nearest to the load is adjusted to make the real part of The admittance at the points CD equals to the characteristic conductance of the line, in absence of the second stub. The stub AB is then adjusted to produce zero susceptance at the point CD.

For triple stubs → 0.125λ apart)

$$\boxed{\begin{array}{l} l_{t_1} = 0.345\lambda \\ l_{t_2} = 0.11\lambda \end{array}}$$